



Lesson 7.8 — Inverse Trigonometric Functions

Big idea: Sin, cos, and tan aren't one-to-one — they fail the horizontal line test. To define inverses, we restrict each to a specific principal interval.

Quick review

arcsin(x) (or $\sin^{-1}$): domain $[-1, 1]$, range $[-\pi/2, \pi/2]$.

arccos(x) (or $\cos^{-1}$): domain $[-1, 1]$, range $[0, \pi]$.

arctan(x) (or $\tan^{-1}$): domain $\mathbb{R}$, range $(-\pi/2, \pi/2)$.

Reading: $\arcsin(x) = \theta$ means $\sin \theta = x$ AND θ is in the principal range.

Calculator caveat: the calculator returns the principal value; you may need to add multiples of 2π or use symmetry to get all angles satisfying an equation.

Worked example

Worked example. Find $\arcsin(\sqrt{3}/2)$ and $\arccos(-1/2)$.

$\arcsin(\sqrt{3}/2) = \pi/3$ (the angle in $[-\pi/2, \pi/2]$ whose sine is $\sqrt{3}/2$).

$\arccos(-1/2) = 2\pi/3$ (the angle in $[0, \pi]$ whose cosine is $-1/2$).

Warm-ups

1. Find $\arcsin(0)$.
2. Find $\arccos(1)$.
3. Find $\arctan(1)$.

Practice

1. Find $\arcsin(-1/2)$.
2. Find $\arccos(0)$.
3. Find $\arctan(-\sqrt{3})$.

Challenge

1. Evaluate $\sin(\arccos(3/5))$. (Hint: draw a right triangle with $\cos = 3/5$.)
2. Solve $\cos \theta = 1/2$ for $0 \leq \theta < 2\pi$.



Answer key — Lesson 7.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 0.
2. 0.
3. $\pi/4$.

Practice

1. $-\pi/6$ (in $[-\pi/2, \pi/2]$).
2. $\pi/2$.
3. $-\pi/3$ (in $(-\pi/2, \pi/2)$).

Challenge

1. Right triangle with adjacent 3, hypotenuse 5 $\Rightarrow$ opposite = 4. $\sin = 4/5$.
2. $\theta = \pi/3$ (Q1) and $5\pi/3$ (Q4).